MATHEMATICS — CLASS XI

Chapter 1: Complex Numbers

Simple, exam-focused notes • Definitions • Formulae • Solved examples • Practice questions

How to use these notes: Read each definition, learn the formula, then try the practice questions without looking at the answers. These notes cover the standard Class 11 Complex Numbers topics; compare the headings with your current textbook/syllabus.

1. What is a complex number?

A **complex number** is a number written in the form **z = a + bi**, where **a** and **b** are real numbers and **i** is the imaginary unit.

$$z = a + bi$$

a is called the real part and **b** is called the imaginary part's coefficient.

$$\text{Re}(z) = a,\ \text{Im}(z) = b$$

Example: If z = 4 – 3i, then Re(z) = 4 and Im(z) = –3. Remember: Im(z) is the coefficient of i, not –3i.

Types of complex numbers

• **Purely real:** b = 0, e.g. 5 = 5 + 0i.
• **Purely imaginary:** a = 0 and b ≠ 0, e.g. –2i.
• **General complex number:** both real and imaginary parts may be non-zero, e.g. 3 + 2i.

2. The imaginary unit i

The imaginary unit is defined by:

$$i = \sqrt{(-1)},\ i^2 = -1$$

Powers of i repeat in a cycle of four:

Power	Value
i■	1
i^1	i
i^2	–1
i^3	–i
i■	1
i■	i
i■	–1
i■	–i

Shortcut: To simplify i■, divide n by 4 and use the remainder: remainder 0 → 1; 1 → i; 2 → –1; 3 → –i.

Example: i^{27}. Since $27 = 4 \times 6 + 3$, the remainder is 3, so $i^{27} = i^3 = -i$.

3. Equality of complex numbers

Two complex numbers are equal only when their real parts are equal and their imaginary coefficients are equal.

If $a + bi = c + di$, then $a = c$ and $b = d$.

Example: If $2x + (y - 1)i = 6 + 3i$, compare parts: $2x = 6$, so $x = 3$; $y - 1 = 3$, so $y = 4$.

4. Addition and subtraction

Add or subtract the real parts together and the coefficients of i together.

$(a + bi) + (c + di) = (a + c) + (b + d)i$

$(a + bi) - (c + di) = (a - c) + (b - d)i$

Example 1: $(3 + 2i) + (5 - 7i) = 8 - 5i$.

Example 2: $(6 + 4i) - (2 - 3i) = 4 + 7i$. Be careful: subtracting $(-3i)$ gives $+3i$.

5. Multiplication

Multiply as usual, then replace every i^2 with -1.

$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$

Example: $(2 + 3i)(4 - i) = 8 - 2i + 12i - 3i^2 = 11 + 10i$, because $i^2 = -1$.

Example: $(1 + 2i)^2 = 1 + 4i + 4i^2 = -3 + 4i$.

6. Complex conjugate

The **conjugate** of $z = a + bi$ is written $z\blacksquare$ (or z^*) and is obtained by changing the sign between the real and imaginary parts.

If $z = a + bi$, then $z\blacksquare = a - bi$.

Example: If $z = 5 - 2i$, then $z\blacksquare = 5 + 2i$.

Useful identities:

$z + z\blacksquare = 2a$

$z - z\blacksquare = 2bi$

$z\, z\blacksquare = a^2 + b^2$

A complex number multiplied by its conjugate is always a non-negative real number.

7. Division of complex numbers

To divide complex numbers, multiply the numerator and denominator by the conjugate of the denominator. This makes the denominator real.

$(a + bi)/(c + di) = [(a + bi)(c - di)]/(c^2 + d^2)$, provided $c^2 + d^2 \neq 0$.

Example: $(3 + 2i)/(1 - i)$. Multiply top and bottom by $(1 + i)$:

$[(3 + 2i)(1 + i)]/[(1 - i)(1 + i)] = (1 + 5i)/2 = 1/2 + (5/2)i$.

Check: $(3 + 2i)(1 + i) = 3 + 3i + 2i + 2i^2 = 1 + 5i$, and $1 - i^2 = 2$.

8. Modulus (absolute value)

For z = a + bi, the **modulus** is the distance of the point (a, b) from the origin in the complex plane.

$$|z| = \sqrt{(a^2 + b^2)}$$

Example: If z = 3 + 4i, then $|z| = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

Important: $|z|^2 = z\,z\blacksquare = a^2 + b^2$.

9. Argand plane / geometrical representation

A complex number z = a + bi can be represented by the point (a, b):

• Horizontal axis: real axis.
• Vertical axis: imaginary axis.
• z = a + bi is plotted at (a, b).

Example: 2 + 3i is represented by the point (2, 3). Its real coordinate is 2 and its imaginary coordinate is 3.

10. Useful formula sheet

Topic	Formula / rule
Imaginary unit	$i^2 = -1$
Powers of i	Repeat every 4 powers
Standard form	z = a + bi
Real and imaginary parts	Re(z) = a; Im(z) = b
Conjugate	$z\blacksquare = a - bi$
Modulus	$\|z\| = \sqrt{(a^2 + b^2)}$
Conjugate product	$z\,z\blacksquare = \|z\|^2 = a^2 + b^2$
Addition	(a + c) + (b + d)i
Multiplication	(ac – bd) + (ad + bc)i

11. Practice questions

A. Simplify the powers of i
1) $i^{1}\blacksquare$ 2) i^{21} 3) $i^{3}\blacksquare$ 4) $i\blacksquare^{3}$

B. Find real and imaginary parts
5) z = 7 – 4i 6) z = –3 + 8i

C. Calculate
7) (2 + 5i) + (4 – 3i)
8) (8 – i) – (3 + 6i)
9) (3 + i)(2 – 4i)
10) $(1 - 3i)^2$

D. Conjugate and modulus
11) Find the conjugate of –4 – 7i.
12) Find |6 + 8i|.
13) Find $z\,z\blacksquare$ if z = 2 – 5i.

E. Division and equality
14) Simplify (2 + i)/(1 + i).
15) If 3x + (y + 2)i = 12 – 5i, find x and y.

12. Answers to practice questions

1) –1 2) i 3) 1 4) –i

5) Re(z) = 7, Im(z) = –4 6) Re(z) = –3, Im(z) = 8

7) 6 + 2i 8) 5 – 7i 9) 10 – 10i 10) –8 – 6i

11) –4 + 7i 12) 10 13) 29

14) (3 – i)/2 = 3/2 – i/2 15) x = 4, y = –7

Exam tips

1. Always write complex numbers in a + bi form at the end.
2. Remember $i^2 = -1$; this is the most common source of mistakes.
3. For division, use the conjugate of the denominator.
4. For equality, compare real parts separately from imaginary coefficients.
5. For i■, divide the exponent by 4 and use the remainder.

End of Chapter 1 Notes